

COVID-19 CT Scan Classification using ShuffleNet V2

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Abstract—The rapid global spread of the Coronavirus Disease 2019 has necessitated the development of fast, accurate, and scalable diagnostic tools. While clinical biochemical tests are the standard, they can suffer from sensitivity issues and long turnaround times. Computed Tomography scans offer detailed visual evidence of pulmonary infections but require time-intensive expert analysis. This paper proposes a highly efficient, automated deep learning framework for classifying coronavirus using Computed Tomography scan imagery. Utilizing the lightweight ShuffleNet V2 architecture implemented in PyTorch, the proposed model categorizes scans into three classes: Covid, Healthy, and Others. By leveraging transfer learning, the model overcomes the limitations of small medical datasets. Experimental results over a 70-15-15 dataset split yield a training accuracy of 95.27 percent, evaluated alongside precision, recall, F1-score, and confusion matrix metrics. The findings demonstrate that computationally lightweight networks can achieve high diagnostic accuracy, making them suitable for rapid point-of-care deployment. Future work aims to enhance generalization using larger datasets and deploy the model as a clinical web application.

Index Terms—COVID-19, Deep Learning, ShuffleNet V2, CT Scan, Image Classification, Transfer Learning, PyTorch

I. INTRODUCTION

THE Coronavirus Disease 2019 (COVID-19), caused by the SARS-CoV-2 virus, has profoundly impacted global healthcare systems. A critical component in controlling the spread of the virus is the ability to screen and diagnose patients rapidly and accurately. To complement biochemical testing, radiological imaging, specifically chest Computed Tomography (CT), has emerged as a vital diagnostic tool. CT scans provide a comprehensive view of the lungs, allowing radiologists to identify characteristic manifestations of COVID-19, such as bilateral, multi-lobe ground-glass opacities and consolidations.

During pandemic peaks, the sheer volume of generated medical imagery far exceeds the capacity of available medical professionals, leading to delayed diagnoses. To break through this diagnostic bottleneck, Artificial Intelligence and deep learning have been widely adopted to automate medical image analysis. While numerous deep learning models have achieved state-of-the-art accuracy in detecting COVID-19, many rely on massive, computationally expensive architectures. These heavyweight models are difficult to deploy on standard clinical workstations or mobile point-of-care devices.

To address the need for a highly accurate yet computationally inexpensive diagnostic system, this project implements the ShuffleNet V2 deep learning architecture. Developed using PyTorch and leveraging transfer learning, the proposed framework classifies CT images into three distinct categories: Covid, Healthy, and Others. The remainder of this paper is structured

as follows: Section II reviews related literature; Section III details the methodology and dataset; Section IV presents the experiments; Section V discusses the results; and Section VI outlines the conclusion.

II. RELATED WORK

Since the onset of the pandemic, extensive research has explored deep learning approaches for the automated detection of COVID-19 from medical imagery. A systematic literature review by Ritika Mahajan, Amritpal Singh, and Aman Singh highlights that early diagnostic systems relied heavily on well-established Convolutional Neural Networks [1]. For instance, Aayush Jaiswal, Neha Gianchandani, Dilbag Singh, Vijay Kumar, and Manjit Kaur utilized a DenseNet201-based deep transfer learning model, achieving a classification accuracy of 96.25 percent [2]. Similarly, Md. Zahangir Alom, M. M. Shaifur Rahman, Mst Nasrin, Tarek Taha, and Vijayan Asari deployed an Inception Recurrent Residual Convolutional Neural Network with transfer learning, achieving a detection accuracy of 98.78 percent [3]. Ophir Gozes *et al.* employed a pre-trained ResNet-50 architecture to analyze CT scans, reporting a sensitivity of 94 percent [4].

Recognizing the need for faster, more accessible systems, researchers began exploring lightweight architectures. Xin Li, Chengyin Li, and Dongxiao Zhu proposed a portable diagnostic system utilizing DenseNet-121 to automatically classify COVID-19, achieving an 88 percent accuracy [5]. Debabrata Dansana *et al.* built a multi-class classifier using VGG-16, successfully distinguishing between COVID-19, pneumonia, and healthy cases [6].

In the realm of highly efficient mobile architectures, the ShuffleNet family has proven to be an exceptional baseline for medical imaging. Cosku Oksuz, Oguzhan Urhan, and Mehmet Kemal Gullu incorporated ShuffleNet into an end-to-end framework combining SqueezeNet and EfficientNet-B0 to extract features at different abstraction levels, noting ShuffleNet's specific design advantages for mobile devices [7]. Runwen Hu *et al.* specifically utilized a model with ShuffleNet V2 as the backbone for automated diagnosis from chest CTs, achieving an accuracy of 91.21 percent through extensive data augmentation [8].

Recent studies have also evaluated the reliability of ShuffleNet V2. Alexander Wong, Xiaoyu Wang, and Andrew Hryniewski explored trust quantification metrics for deep neural networks, demonstrating that ShuffleNet V2 achieves a strong "NetTrustScore" comparable to massive architectures like ResNet-50 while utilizing only 1.32 million parameters [9]. Furthermore, Shravan Venkatraman, Pavan Kumar S,

Rakesh Raj Madavan, and Chandrakala S utilized ShuffleNet V2 as a benchmark for Uncertainty-Guided Progressive Learning frameworks [10]. Building upon this foundation, our work directly applies ShuffleNet V2 to a multi-class dataset to achieve highly accurate COVID-19 classification.

III. METHODOLOGY

A. Dataset Description and Preprocessing

The foundation of this study is a COVID Multiclass Dataset of CT Scans designed to train the neural network to differentiate between specific pulmonary conditions. The dataset is annotated into three distinct classes:

- **Covid:** Scans presenting characteristic viral infections.
- **Healthy:** Normal lung scans showing no signs of infection.
- **Others:** Scans presenting signs of non-COVID viral or bacterial pneumonia.

To ensure robust model training and to strictly evaluate the model's ability to generalize to unseen data, the dataset was rigorously divided into three subsets: 70 percent of the data was utilized for Training, 15 percent was reserved for Validation during the training phase, and the remaining 15 percent was isolated for final Testing.

B. Proposed Architecture: ShuffleNet V2

The core of our classification framework is the ShuffleNet V2 architecture. Traditional Convolutional Neural Networks require massive computational resources due to dense convolution operations. ShuffleNet V2 overcomes this by utilizing point-wise grouped convolutions and a novel channel shuffle operation. This ensures that the inputs for the next layer are gathered from different groups, allowing seamless information exchange across the entire feature map while keeping the mathematical and computational overhead incredibly low.

C. Transfer Learning Paradigm

A significant challenge in medical image classification is the scarcity of vast datasets required to train deep neural networks from scratch. To combat this, our methodology employs transfer learning. The ShuffleNet V2 model was initialized with pre-trained weights. The lower layers of the network, which act as generalized feature extractors, were retained. The final fully connected layers were modified and fine-tuned specifically to map the extracted deep features to our three specific clinical classes.

D. Implementation Details

The entire pipeline was developed using the Python programming language. The ShuffleNet V2 architecture and training loops were implemented using PyTorch, a highly flexible deep learning framework. Due to the computational demands of deep learning, the experiments were hosted and executed on Google Colab, utilizing cloud-based GPU acceleration. Additional data processing, statistical analysis, and performance visualizations were handled using Scikit-learn and Matplotlib.

IV. EXPERIMENTS

To provide a comprehensive evaluation of the ShuffleNet V2 model's diagnostic capabilities, we measured its performance using multiple statistical metrics. Given that medical datasets can often be imbalanced, relying solely on accuracy can yield misleading results. Thus, the following metrics were utilized:

- **Accuracy:** The ratio of correctly predicted observations to the total observations.
- **Precision:** The ratio of correctly predicted positive observations to the total predicted positive observations.
- **Recall:** The ratio of correctly predicted positive observations to all actual observations in the actual class.
- **F1-Score:** The harmonic mean of Precision and Recall.
- **AUC-ROC:** The Area Under the Receiver Operating Characteristic Curve.
- **Confusion Matrix:** A tabular layout that visualizes the performance across the three classes.

Upon completion of the training cycle over the 70-15-15 dataset split, the model achieved a notable Training Accuracy of 95.27 percent.

V. DISCUSSION AND SUMMARY

The training accuracy of 95.27 percent achieved by our implementation strongly supports the hypothesis that lightweight neural networks are highly capable of performing complex medical imaging tasks. When comparing these results to the broader literature, our implementation demonstrates competitive performance. For example, Runwen Hu *et al.* utilized a similar ShuffleNet V2 backbone combined with data augmentation, reporting an accuracy of 91.21 percent [8].

While heavyweight architectures like ResNet-50 may occasionally offer marginal improvements in absolute classification accuracy, they require significantly more memory and inference time. In a real-world clinical setting, a radiologist requires rapid diagnostic feedback. As demonstrated by Alexander Wong, Xiaoyu Wang, and Andrew Hryniewski [9], lightweight networks like ShuffleNet V2 require a fraction of the computational footprint while maintaining strong trust quantification scores. This proves them to be reliable automated assistants that can be run on basic hospital computer terminals without the need for expensive dedicated servers.

VI. CONCLUSION AND FUTURE WORK

This paper presented an automated deep learning framework for the multiclass classification of COVID-19 from Computed Tomography scan images. By implementing the lightweight ShuffleNet V2 architecture using PyTorch and leveraging transfer learning, the model successfully distinguished between Covid, Healthy, and Other pulmonary classes. Evaluating the model over a 70-15-15 dataset split yielded a highly impressive training accuracy of 95.27 percent.

Looking forward, the future work for this project is organized into three primary objectives. First, we aim to further improve overall accuracy through advanced optimization. Second, we plan to validate the model by training and testing it on significantly larger global CT datasets to ensure robust generalizability. Finally, the ultimate goal is to deploy the finalized

model as a user-friendly web application, allowing medical professionals to upload CT scans and receive instantaneous diagnostic predictions.

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